SET - A

KENDRIYA VIDYALAYA SANGATHAN, PATNA REGION
PRE- BOARD EXAMINATION (SESSION 2024-25)
CLASS: X
SUBJECT: MATHEMATICS STANDARD (041)

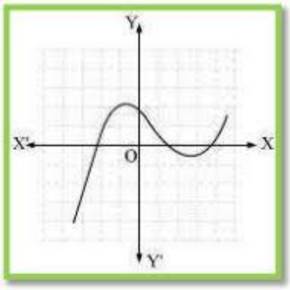
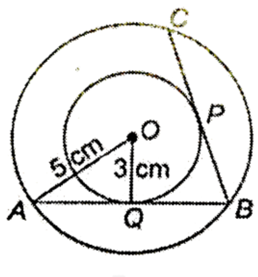
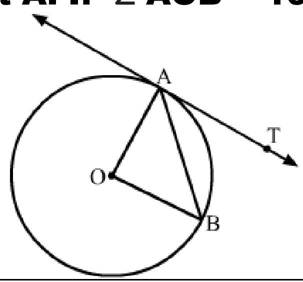
TIME: 3 hours **MAX.MARKS: 80**

General Instructions:

Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take π =22/7 wherever required if not stated.
11. Use of calculators is not allowed.

	SECTION A	
	Section A consists of 20 questions of 1 mark each.	
1.	If A =2n+13, B= n+7 , where n is a natural number , then HCF of A and B is: (a)2 (b) 1 (c) 3 (d) 4	1
2.	The graph of y= f(x) is given below, for some polynomial f(x) . The number of zeroes of f(x) is /are	1

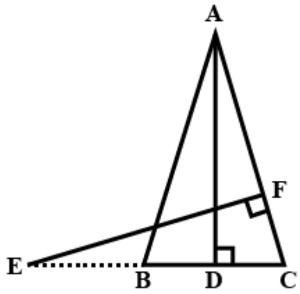
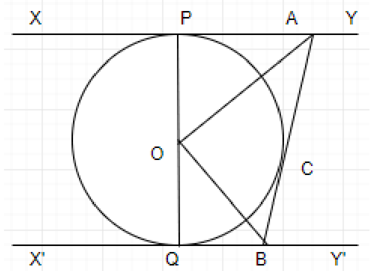
	 **(a)0 (b) 1 (c) 2 (d) 3**	
3.	**The value of k for which the system of equation x + y – 4 =0 and 2x + ky =3 , has no solution, is** (a) **-2** **(b) ≠ 2** **(c) 3** **(d) 2**	1
4.	**If the quadratic equation $x^2 + 4x + k = 0$ has real and distinct roots , then** **(a)k < 4 (b) k > 4 (c) k ≥ 4 (d) k ≤ 4**	1
5.	**What will be the value of P(not E) if P(E) = 0.07.** **(a) 0.07 (b)0.93 (c) 0.73 (d) 1**	1
6.	**In an AP , if the common difference (d) = -4 and the seventh term (a_7) is 4 , then the first term is** **(a)28 (b) 32 (c) 24 (d) 20**	1
7.	**If P ($a/3$, 4) is the mid point of the line segment joining the points Q(-6 , 5) and R (-2 . 3) , then the value of a is** **(a)-4 (b) -6 (c) 12 (d) -12**	1
8.	**Using the empirical formula, find the mode of a distribution whose mean is 8.32 and the median is 8.05.** **(a) 7.51 (b) 8.51 (c) 6.51 (d) 5.51**	1
9.	**In Fig. two concentric circles of radii 3 cm and 5 cm are given . Then length of chord BC which touches the inner circles at P is equal to**  **(a)4 cm (b) 6 cm (c) 8 cm (d) 10 cm**	1
10.	**In given figure , O is the centre of the circle , AB is a chord and AT is the tangent at A. If ∠ AOB = 100° then ∠BAT is**  	1

	(a) 30° (b) 40° (c) 50° (d) 100°	
11.	$9\sec^2 A - 9\tan^2 A$ is equal to (a) 1 (b) 9 (c) 8 (d) 0	1
12.	From a point Q the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is (a) 7 cm (b) 12 cm (c) 15 cm (d) 24.5 cm	1
13.	If $\sin\theta = \cos\theta$, then the value of $\tan^2\theta + \cot^2\theta$ is (a) 2 (b) 4 (c) 1 (d) 10/3	1
14.	An arc of a circle is of length 5π cm and the sector it bounds has an area of $20\pi\ cm^2$. The radius of the circle is (a) 4 cm (b) 10 cm (c) 8 cm (d) 15 cm	1
15.	In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. The area of the sector formed by the arc is: (a) 200 cm^2 (b) 220 cm^2 (c) 231 cm^2 (d) 250 cm^2	1
16.	Roots of the quadratic equation $2x^2 - 4x + 3 = 0$ are (a) real and equal (b) real and distinct (c) not real (d) real	1
17.	A bag contains 5 black balls, 4 white balls and 3 red balls. If a ball is selected random wise, the probability that it is a black or a red ball is (a) 5/12 (b) 2/3 (c) 2/5 (d) 5/7	1
18.	For the distribution Class: 0-5, 5-10, 10-15, 15-20, 20-25 Frequency: 10, 15, 12, 20, 9 The sum of upper limit of the median class and the modal class is (a) 15 (b) 25 (c) 30 (d) 35	1
	DIRECTION: In the question number 19 and 20, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct option (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion(A). (b) Both assertion(A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A) . (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	1
19	Statement A (Assertion) : The curved surface area of a cone of base radius 6 cm and slant height 10 cm is $60\pi\ cm^2$. Statement R(Reason) : Curved surface area of a cone = $\pi r^2 h$, where r be the radius and h be the height of cone.	
20.	Statement A(Assertion) : The points (-2, 3) , (1, 2) and (7, 0) are collinear. Statement R (Reason): If we plot these points on graph they all lie on a line.	1

Table for question 18:

Class	0-5	5-10	10-15	15-20	20-25
Frequency	10	15	12	20	9

	SECTION B	
	Section B consists of 5 questions of 2 marks each.	
21.	**Five cards – the ten, jack, queen, king and ace of diamonds, are well- shuffled with their face downwards. One card is then picked up at random.** **(i) What is the probability that the card is the queen?** **(ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?**	2
22.	**Find the ratio in which line segment joining A(1,-5) and B(-4,5) is divided by the x-axis. Also find the coordinates of the point of division.**	2
23.	**Given 15 cot A= 8, find sin A and sec A.** **OR** **If tan (A+B) = √3 and tan (A-B) = 1/√3; 0° < A+B ≤ 90°; A > B, find A and B .**	2
24.	**Check whether 6^n can end with the digit 0 for any natural number n.**	2
25.	**Find the values of *y* for which the distance between the points P(2, – 3) and Q(10, *y*) is 10 units.** **OR** **Find the coordinates of the point which divides the join of (–1, 7) and (4, –3) in the ratio 2 : 3.**	2
	SECTION C	
	Section C consists of 6 questions of 3 marks each	
26.	**Prove that 3 + 2√5 is an irrational.**	3
27.	**Show that ½ and -3/2 are the zeroes of the polynomial $4x^2 + 4x - 3$ and verify the relationship between zeroes and co-efficient of polynomial.**	3
28.	**A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/hr less, then it would have taken 3 hours more to cover the same distance. Find the speed of the train.** **OR** **Solve the equation-** **1/(x+4) – 1/(x-7) = 11/30, x not equal to -4,7**	3
29.	**Prove that:** $\cos A/(1+\sin A) + (1+\sin A)/\cos A = 2\sec A$	3

30.	**E is a point on side CB produced of an isosceles triangle ABC with AB = AC. If AD ⊥ BC and EF ⊥ AC, prove that ΔABD ~ Δ ECF.**  **OR** **State and prove Basic Proportionality theorem.**	3
31.	**A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115°. Find the total area cleaned at each sweep of the blades.**	3
	SECTION D	
	Section D consists of 4 questions of 5 marks each	
32.	**A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid ` 27 for a book kept for seven days, while Susy paid ` 21 for the book she kept for five days. Find the fixed charge and the charge for each extra day.** **OR** **(i) Solve $2x + 3y = 11$ and $2x - 4y = -24$ and hence find the value of 'm' for which $y = mx + 3$.** **(ii) Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. What are their present ages?**	5 2 ½ 2 ½
33.	**In the given Figure , XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that ∠ AOB = 90°.**  **OR** **Prove that the parallelogram circumscribing a circle is a rhombus.**	5

34.	**Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30°, respectively. Find the height of the poles and the distances of the point from the poles.**	5
35.	**If the median of the distribution given below is 28.5, find the values of X and Y.**	5

Class Interval	Frequency
0-10	5
10-20	X
20-30	20
30-40	15
40-50	Y
50-60	5
Total	60

SECTION E

36. **Your friend Veer wants to participate in a 200 m race. He can currently run that distance in 51 seconds and with each day of practice it takes him 2 seconds less. He wants to do in 31 seconds.**

(i) What is the minimum number of days he needs to practice till his goal is achieved ? 1

(ii) If n^{th} term of an AP is given by $a_n = 2n + 3$, then find common difference of an AP. 1

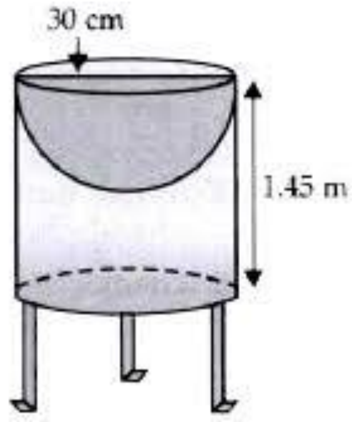
(iii) Find the time taken by Veer to run 200 m on the 6^{th} day of 4

practice. **2**

OR

Find the value of x, for which 2x, x +10 , 3x + 2 are three consecutive terms of an AP.

37. **A bird bath can be garden ornament , small reflecting pool, outdoor sculpture and also can be a part of creating a vital wildlife garden . Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and its radius is 30 cm.** 4



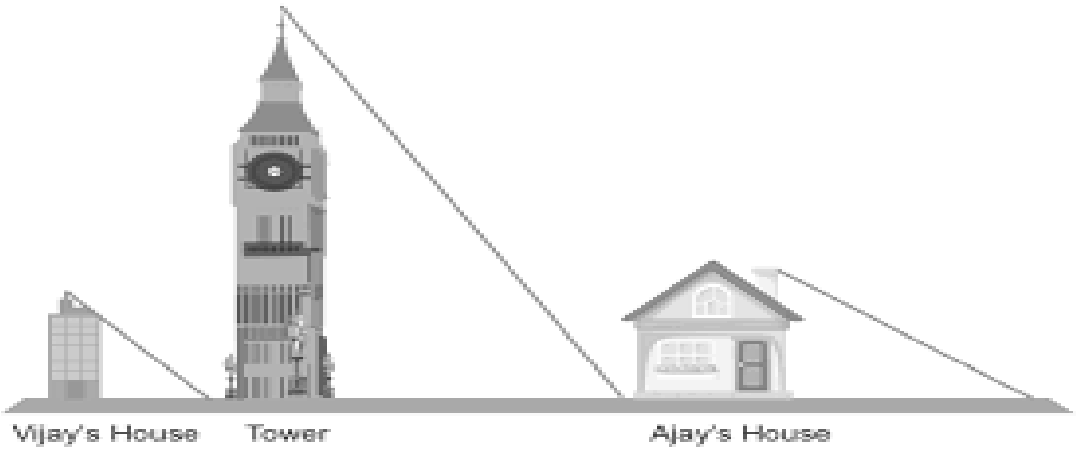
Based on the above information answer the following question:

- **(i) Find the curved surface area of cylinder.** **1**
- **(ii) Find the curved surface area of hemisphere.** **1**
- **(iii) Find the volume of the cylinder .** **2**

OR

Find the total surface area of the bird bath.

38. **Vijay is trying to find the height of a tower near his house. He is using the properties of similar triangles. The height of Vijay's house is 20 m and casts a shadow of length 10 m at the ground. At the same time, the tower casts 50 m long shadow on the ground and the house of Ajay casts 20 m shadow on the ground.**
Based on the above information answer the following question:



	(i) What is the height of the tower? 1 (ii) What is the height of Ajay's house? 1 (iii) What will be the length of the shadow of the tower when Vijay's house casts a shadow of 12 m? 2 OR What will be the length of the shadow of Ajay's house when the tower casts a shadow 40 m long?	